Annex E

Reticular Local Abstraction, Epistemic CNNs, and the Bridge to the Principle of Computational Equivalence

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September 2025

Abstract

This annex develops the "bridge layer" between the general framework of *Reticular Local Abstraction* (RLA) and *Compact Reticular Computability* (CRC) introduced in Annex A, the engineered *Epistemic Convolutional Neural Networks* (ECNN) formalised in Annex C, and Wolfram's *Principle of Computational Equivalence* (PCE) as explored through one–dimensional Cellular Automata (CA) [Wolfram, 1984, Wolfram, 2002]. The structural gaps addressed here are: (i) an explicit alignment of terminology and axioms with Annex A (IOA, EC, TC and CRC), (ii) a precise reticular reading of ECNN architectures and their epistemic heads, and (iii) a clearer, diagnostics–driven mapping from compact reticula to CA dynamics, with falsifiable conjectures rather than ontological claims.

RLA represents systems as finite reticula of levels of abstraction linked by transmissions that may be injective, quasi–injective (on Turing–critical sets), or non–injective (information collapse). CRC characterises when such reticula are internally ontologically autonomous, epistemically closed and fully Turing–computable [Church, 1936, Turing, 1936/1937]. ECNNs re–interpret CNN stacks as reticula and add an epistemic channel that outputs structured *unknown* and *contradiction* signals, evaluated through Popper–style challenge sets [Popper, 1959, Goodfellow et al., 2016]. This annex introduces a systematic dictionary between compact reticula and one–dimensional CA, and proposes that reticula which are both Turing–like and empirically irreducible, with non–trivial emergence, correspond (under an explicit encoding) to complex (class IV/V) CA regimes, and conversely [Li & Packard, 1990, Cook, 2004, Wolfram, 2002, Shalizi & Crutchfield, 2001].

The bridge is deliberately epistemic rather than ontological: it provides a shared computational vocabulary and falsifiable diagnostics for relating layered computation, logical limits and canonical models of complexity, without committing to claims about "what the universe really is".

Contents

1 Introduction

Scientific practice across physics, biology and machine learning recurrently encounters two stubborn motifs:

1. *persistent macro–regularities* that resist transparent reduction to micro–level predicates, even when micro–dynamics are well understood [Anderson, 1972, Noble, 2012];

2. *logical limits* that emerge as soon as a substrate attains Turing–equivalent computational power [Turing, 1936/1937, Rice, 1953, Chaitin, 2005].

Reticular Local Abstraction (RLA) and Compact Reticular Computability (CRC), introduced in Annex A, offer a unified vocabulary for both issues by describing models as lattices of levels linked by transmissions whose injectivity controls how information, undecidability and emergent properties flow across scales.

Epistemic Convolutional Neural Networks (ECNN), formalised in Annex C, instantiate this viewpoint on contemporary deep architectures [Goodfellow et al., 2016]. Convolutional and pooling layers are reinterpreted as levels and transmissions; non–injective pooling implements controlled information collapse; and an epistemic head produces explicit *unknown* and *contradiction* outputs. These outputs are evaluated with Popper–style challenge suites that target epistemic behaviour, not just predictive accuracy [Popper, 1959]).

Wolfram's Principle of Computational Equivalence (PCE) suggests that, beyond a low threshold, many natural and artificial systems achieve a form of computational universality and irreducible behaviour [Wolfram, 1984, Wolfram, 2002]. One–dimensional cellular automata (CA) are central to this proposal: simple local rules give rise to a spectrum of dynamics, from fixed patterns (class I) to complex structures (class IV/V) [Li & Packard, 1990, Cook, 2004].

This annex has three aims:

1. to summarise and specialise the RLA/CRC axioms of Annex A to the neural and epistemic architectures of Annex C;
2. to articulate ECNNs as compact reticula with an epistemic interface and falsification protocol;
3. to construct a bridge between compact reticula and one–dimensional CA, and to formulate conjectures linking structural diagnostics to CA dynamical classes in the spirit of PCE [Wolfram, 1984, Wolfram, 2002].

The treatment is intentionally conceptual and methodological. No empirical dataset is assumed, and no claim is made about the actual dynamical class of any particular trained network. All constructions are *epistemic*: they formalise how we model and reason about systems, not an ultimate ontology of the systems themselves.

2 Background and Alignment with Annex A & C

2.1 Reticular Local Abstraction and CRC (from Annex A)

We briefly recall and specialise the core notions of Annex A (cf. [Church, 1936, Turing, 1936/1937, Rice, 1953]).

Definition 2.1 (Level of abstraction and RLA reticulum)**.** *A* level of abstraction *(or simply* level*) is a pair $L = \langle D(L), \Sigma(L)\rangle$, where:*

- *$D(L)$ is a set (or discretisable space) of admissible states;*
- *$\Sigma(L)$ is a set of effective rules acting on $D(L)$ (dynamical laws, update equations, inference rules, programs).*

An RLA reticulum *is a finite sequence*

$$R = (L_1, \dots, L_n)$$

together with transmissions $\tau_{i\to i+1} : D(L_i) \to D(L_{i+1})$ *for* $i = 1, \dots, n-1$.

Definition 2.2 (Transmission classes)**.** *Let $S \subseteq D(L_i)$ be a subset of interest. A transmission $\tau_{i\to j}$ is:*

- injective on S *if* $x_1, x_2 \in S$ *and* $\tau_{i\to j}(x_1) = \tau_{i\to j}(x_2)$ *imply* $x_1 = x_2$*;*

- quasi–injective on S *if there exists a partition* $S = S' \cup S''$ *such that* $\tau_{i\to j}$ *is injective on* S' *and* S'' *is declared irrelevant for a given analysis (e.g. below resolution);*

- non–injective on S *if there exist distinct* $x_1, x_2 \in S$ *with* $\tau_{i\to j}(x_1) = \tau_{i\to j}(x_2)$ *and the distinction between* x_1, x_2 *is relevant for at least one family of properties.*

Definition 2.3 (Turing–like level)**.** *A level* $L_i = \langle D(L_i), \Sigma(L_i)\rangle$ *is* Turing–like *if the formalism represented by* $\Sigma(L_i)$ *can, up to a computable encoding of inputs and outputs, simulate a universal Turing machine (or an equivalent model of computation) [Church, 1936, Turing, 1936/1937].*

Definition 2.4 (Epistemic horizon)**.** *An* epistemic horizon $\mathcal{H}$ *for a modelling practice is a specified set of observables, questions and explanatory tasks that are considered in scope for a given reticulum R. Formally,* $\mathcal{H}$ *is a finite or countable family of predicates and queries about the domain, together with a specification of which answers or explanations are admissible.*

Definition 2.5 (IOA, EC, TC and compact reticulum)**.** *An RLA reticulum R is:*

- internally ontologically autonomous (IOA) *if all explanatory entities and laws for the domain under horizon* $\mathcal{H}$ *are represented internally by levels and transmissions;*

- epistemically closed (EC) *with respect to* $\mathcal{H}$ *if every observable or question in* $\mathcal{H}$ *can be represented and explained using only the states and rules of R;*

- Turing–computable (TC) *if all* $\Sigma(L_i)$ *and* $\tau_{i\to j}$ *are effectively realisable by Turing machines.*

R is a compact reticulum *if it satisfies IOA, EC and TC for some specified horizon* $\mathcal{H}$*.*

Definition 2.6 (Compact Reticular Computability (CRC))**.** *A compact reticulum R exhibits* Compact Reticular Computability *if:*

1. *a Turing machine* T *can simulate every* $\Sigma(L_i)$ *and* $\tau_{i\to j}$ *to arbitrary precision;*

2. *at least one level* L_k *is Turing–like;*

3. *there exist macro–properties at some higher level* L_j*,* $j > k$*, that are computable with respect to* $\Sigma(L_j)$ *but not definable, in the languages of lower levels, as predicates on* $D(L_i)$ *for* $i < j$*.*

CRC thus extends classical computability from single machines to multi–level epistemic architectures, without altering any existing result of computability theory [Rice, 1953, Chaitin, 2005].

2.2 Working axioms (specialised form)

We specialise the Annex A axioms to the setting where at least one level is Turing–like and we care about how limits and macro–structure propagate [Rice, 1953].

Axiom 2.1 (Undecidability at Turing–like levels)**.** *Let* L_i *be Turing–like and* $C_i \subseteq D(L_i)$ *a critical set on which* $\Sigma(L_i)$ *is Turing–complete. Then any non–trivial extensional property* $U \subseteq C_i$ *is undecidable with respect to* $\Sigma(L_i)$*, in the sense of Rice's theorem [Rice, 1953].*

Axiom 2.2 (Propagation of limits through quasi–injective maps). *If $\tau_{i\to j}$ is effectively computable and quasi–injective on a critical subset $C_i \subseteq D(L_i)$, then undecidable properties on C_i cannot become decidable on $\tau_{i\to j}(C_i)$ by post–processing.*

Axiom 2.3 (Emergence via non–injective collapse). *If $\tau_{i\to j}$ is non–injective on a subset $S \subseteq D(L_i)$ relevant for a family of properties of interest, then there exists at least one macro–property on $D(L_j)$ that is not definable as a predicate on $D(L_i)$ alone. Such a property is called* emergent *relative to L_i [Anderson, 1972, Noble, 2012].*

These axioms make explicit the two key structural tensions: quasi–injective links preserve logical limits, while non–injective links trade away micro–information to gain new macro–structure.

2.3 Epistemic CNNs as compact reticula (from Annex C)

Annex C defines ECNNs as engineered instances of compact reticula leveraging standard deep learning [Goodfellow et al., 2016].

Definition 2.7 (ECNN (informal recap)). *An* Epistemic Convolutional Neural Network *(ECNN) is a tuple*

$$\mathcal{E} = (L_0, \ldots, L_n, L_{\mathrm{ep}}, \Sigma, M)$$

where:

- *$L_0, \ldots, L_n$ are levels corresponding to input, convolution, pooling, dense and representation blocks of a CNN;*
- *L_{ep} is an epistemic level carrying structured epistemic states;*
- *Σ is a finite set of Turing–computable update rules and transmissions;*
- *M is an epistemic matrix encoding concepts, rules, coherence constraints and policies.*

The representation channel $B_\theta : \mathcal{X} \to \mathbb{R}^d$ maps inputs to high–level embeddings. An Epistemic Computational Unit (ECU) and an epistemic head H_ψ map (z, M) into epistemic artefacts and final distributions over labels and epistemic states $\mathcal{Y} \cup \{\perp, \perp_c\}$ (label, unknown, contradiction).

Under the IOA/EC/TC conditions of Annex A and the explicit design constraints of Annex C, an ECNN becomes a compact reticulum, and—if it contains a Turing–like level with emergent macro–properties—an instance of CRC.

3 Reticular View of ECNN Architectures

3.1 Representation channel and transmissions

Let $\mathcal{X}$ be the input space and $B_\theta : \mathcal{X} \to \mathbb{R}^d$ the CNN–based representation channel. We view each layer as a level $L_i = \langle D(L_i), \Sigma(L_i)\rangle$, where $D(L_i)$ is the activation space and $\Sigma(L_i)$ is the rule that computes activations from predecessors [Goodfellow et al., 2016].

- **Convolutional levels**: $D(L_c) = \mathbb{R}^{H\times W\times C}$; transmissions $\tau_{c\to c+1}$ are local linear operators followed by non–linearities. On discriminative subsets and up to symmetries, they are quasi–injective.

- **Pooling levels**: $D(L_p)$ share shape with their inputs; transmissions $\tau_{p\to p+1}$ implement max or average pooling over neighbourhoods, and are non–injective on relevant states.

- **Dense levels**: $D(L_d) = \mathbb{R}^m$; transmissions are affine maps plus non–linearities, often many–to–one in practice.

Residual or skip connections can be encoded as additional transmissions that bypass collapse, thereby preserving quasi–injective paths on Turing–critical subsets.

3.2 Epistemic head and Popper–χ challenge suites

On top of the representation channel we place the ECU and epistemic head (Annex C). Given $z = B_\theta(x)$ and epistemic matrix M, the ECU produces an epistemic artefact a containing:

- a primary decision $y^\star \in \mathcal{Y} \cup \{\perp, \perp_c\}$;

- optional confidence scores and auxiliary flags;

- a structured explanation or rationale constrained by M.

The epistemic head H_ψ maps a to a distribution over $\mathcal{Y} \cup \{\perp, \perp_c\}$.

Definition 3.1 (Popper–χ challenge suites)**.** *A* Popper–χ challenge suite χ *for an ECNN consists of three families of test cases:*

- ***Contradictory pairs*** Ω_{CP}*: pairs* (x, x') *such that assigning labels* (y, y') *with* $y = y'$ *would violate constraints in* M *(e.g. mutual exclusivity, temporal incoherence);*

- ***Undecidability triggers*** Ω_{UT}*: inputs for which, relative to* M*, no label in* $\mathcal{Y}$ *is justifiable; the epistemically correct response is* $\perp$ *[Popper, 1959];*

- ***Semantic adversaries*** Ω_{SA}*: inputs that preserve low–level statistics but invert or scramble semantic content, testing robustness and invariances [Goodfellow et al., 2016].*

Definition 3.2 (Core epistemic metrics)**.** *Given an ECNN* F *and a challenge suite* χ*, define:*

$$\mathrm{CR} = \frac{\#\{(x, x') \in \Omega_{\mathrm{CP}} : F(x), F(x') \textit{ jointly violate } M \textit{ and no } \perp_c \textit{ is emitted}\}}{\#\Omega_{\mathrm{CP}}},$$

$$\mathrm{IR} = \frac{\#\{x \in \Omega_{\mathrm{UT}} : F(x) \textit{ emits } \perp\}}{\#\Omega_{\mathrm{UT}}}.$$

CR *is the* contradiction rate*;* IR *is the* indecidability recall*.*

Epistemically aligned ECNNs aim for low CR and high IR, in addition to good classical metrics (accuracy, calibration, robustness) [Goodfellow et al., 2016].

3.3 Reticular diagnostics for ECNNs

We now specialise the generic diagnostics of Annex A [Shalizi & Crutchfield, 2001, Wolfram, 2002] to ECNNs.

Definition 3.3 (Coefficient of Collapse CC in ECNNs)**.** *Let $\tau_{i\to j}$ be a layer–to–layer transmission and $S \subseteq D(L_i)$ a finite sample of activations. The* coefficient of collapse *on S is*

$$CC_{i\to j}^{(\mathrm{ECNN})} = 1 - \frac{|\tau_{i\to j}(S)|}{|S|} \in [0,1],$$

where $\tau_{i\to j}(S)$ is counted up to a chosen numerical tolerance.

Definition 3.4 (Index of Emergence IE in ECNNs)**.** *Let $\mathcal{Q}$ be a family of candidate macro–properties on $D(L_j)$ (e.g. linear decision boundaries, clustering structure, simple symbolic descriptors). The* index of emergence *from L_i to L_j is*

$$IE_{i\to j}^{(\mathrm{ECNN})} = \frac{\#\{Q \in \mathcal{Q} : Q \textit{ is computable at } L_j \textit{ and not definable on } D(L_i)\}}{|\mathcal{Q}|}.$$

In practice, non–definability is tested with approximations (linear probes, symbolic regression, restricted hypothesis classes). A non–zero $IE_{i\to j}^{(\mathrm{ECNN})}$ indicates that, within the chosen probe family, some macro–properties at L_j cannot be reduced to predicates on $D(L_i)$.

Definition 3.5 (Reticular irreducibility index RIR in ECNNs)**.** *Let G_t be the causal graph induced by layer activations and transmissions up to time t (for recurrent or iterative networks), with vertex set $V(G_t)$. Let* $\mathrm{MinCut}(G_t)$ *be the smallest number of vertices whose removal disconnects inputs from outputs over that horizon. For a finite horizon T, the* reticular irreducibility index *is*

$$RIR^{(\mathrm{ECNN})} = 1 - \frac{1}{T}\sum_{t=1}^{T} \frac{\mathrm{MinCut}(G_t)}{|V(G_t)|}.$$

Low $RIR^{(\mathrm{ECNN})}$ indicates that small cuts capture most causal structure (high compressibility); high $RIR^{(\mathrm{ECNN})}$ indicates that no small cut suffices, serving as a proxy for computational irreducibility [Shalizi & Crutchfield, 2001, Wolfram, 2002]).

4 Cellular Automata and the RLA–CA Bridge

4.1 One–dimensional cellular automata

A one–dimensional cellular automaton (CA) is defined by [Wolfram, 1984, Li & Packard, 1990]:

- a finite alphabet $\mathcal{A}$,
- a configuration space $\mathcal{A}^{\mathbb{Z}}$ or $\mathcal{A}^{\{1,\dots,N\}}$,
- a local update rule $f : \mathcal{A}^{2r+1} \to \mathcal{A}$ of finite radius r,
- a global map $F : \mathcal{A}^{\mathbb{Z}} \to \mathcal{A}^{\mathbb{Z}}$ defined by $(F(c))_i = f(c_{i-r}, \dots, c_{i+r})$.

Wolfram's empirical classification distinguishes four qualitative classes: (I) fixed point, (II) periodic, (III) chaotic, (IV) complex [Wolfram, 1984, Wolfram, 2002]. Certain elementary CA (e.g. rule 110) are known to be Turing–complete [Cook, 2004].

4.2 Dictionary between reticula and CA

We outline the structural dictionary that underpins the bridge [Li & Packard, 1990, Wolfram, 2002]:

- **Levels and tapes**: RLA levels correspond to abstractions over CA configurations; conversely, CA tapes can be seen as special cases of level states with discrete dynamics.
- **Transmissions and local rules**: transmissions $\tau_{i\to j}$ correspond to local update rules and coarse–grainings; injective vs non–injective behaviour maps to information–preserving vs collapsing CA morphisms.
- **Universality and Turing–like levels**: Turing–complete CA rules correspond to Turing–like levels; reticula with Turing–like levels can, under suitable encodings, be implemented by CA with additional bookkeeping [Cook, 2004].
- **Irreducibility**: high RIR in reticula corresponds to CA regimes where long–range correlations and complex structures resist coarse–grained summarisation, as in class IV/V [Wolfram, 1984, Shalizi & Crutchfield, 2001].

4.3 Constructive encoding Φ: RLA $\to$ CA

Definition 4.1 (Encoding $\Phi : R \to \mathrm{CA}$)**.** *Let R be a compact reticulum. An encoding $\Phi(R)$ is a one–dimensional CA constructed as follows (abstractly):*

1. *Choose finite alphabets that encode states of each level and internal registers.*
2. *Encode joint reticular states at a discrete time t as a CA configuration $c_t \in \mathcal{A}^{\mathbb{Z}}$ via a computable coding γ that respects the product structure of $D(L_i)$.*
3. *Design a local CA rule f such that one global CA step $F(c_t)$ simulates one synchronous update of all levels and transmissions under Σ and $\{\tau_{i\to j}\}$.*

The resulting CA $\Phi(R)$ is such that $F^t(c_0)$ encodes the reticular evolution up to time t, up to computable decoding [Wolfram, 2002, Shalizi & Crutchfield, 2001].

This construction is standard in spirit: finite collections of Turing–computable maps can be simulated by CA with appropriate bookkeeping. What matters here is that quasi–injective and non–injective structure in the reticulum is mirrored by information–preserving and collapsing patterns in the CA dynamics.

4.4 Reverse encoding Ψ: CA $\to$ RLA

Definition 4.2 (Encoding $\Psi : \mathrm{CA} \to R$)**.** *Given a one–dimensional CA with rule f, we define a two–level reticulum:*

- L_{micro} *with states given by CA configurations and rules implementing* F*;*
- L_{macro} *with states given by coarse–grained descriptions of configurations (e.g. block averages, domain decompositions) and rules describing their evolution.*

Transmissions $\tau_{\text{micro}\to\text{macro}}$ *are chosen as coarse–grainings or observables of interest (e.g. majority voting on blocks, feature maps). The resulting reticulum* $\Psi(CA)$ *is compact whenever* F*, the coarse–grainings and the observables are Turing–computable [Li & Packard, 1990, Shalizi & Crutchfield, 2001].*

This reverse mapping shows that CA can be treated as special cases of reticula, and that CA–based complexity can be rephrased in RLA/CRC terms.

5 Conceptual Results and Conjectures

5.1 Propagation versus blocking in ECNN reticula

We first restate, in ECNN form, the structural tension between quasi–injective propagation and non–injective blocking [Rice, 1953, Anderson, 1972].

Proposition 5.1 (Propagation vs blocking of undecidability)**.** *Let* R_{ECNN} *be the reticulum induced by an ECNN and assume it is compact and contains a Turing–like level* L_i*.*

1. *If* $\tau_{i\to i+1}$ *is effectively computable and quasi–injective on a critical subset* $C_i \subseteq D(L_i)$*, then any non–trivial extensional property* $U \subseteq C_i$ *that is undecidable at* L_i *remains undecidable at* L_{i+1}*.*
2. *If* $\tau_{i\to i+1}$ *is non–injective on relevant states, then (i) propagation of some undecidable properties may be blocked, and (ii) by Axiom 2.3 at least one emergent macro–property arises at* L_{i+1}*.*

Sketch. (1) is a Rice–style argument: if U became decidable at L_{i+1}, quasi–injectivity would allow one to decide U at L_i by composing $\tau_{i\to i+1}$ with the hypothetical decider, contradicting Axiom 2.1 and [Rice, 1953]. (2) follows from Axiom 2.3 and the observation that non–injective maps can identify states whose distinct computational behaviour carried the undecidability, thereby potentially blocking it [Anderson, 1972, Noble, 2012]. □

In ECNNs, pooling and bottleneck layers are primary candidates for the second case, whereas carefully designed residual paths can preserve quasi–injective channels for Turing–critical traces.

5.2 Design regimes for ECNN reticula

We distinguish four qualitative design regimes, according to memory capacity and collapse aggressiveness, inspired by both RLA/CRC and CA classifications [Wolfram, 1984, Li & Packard, 1990, Wolfram, 2002].

Regime	Memory	Collapse	Diagnostics (qualitative)
R1	Bounded	Aggressive	Low RIR, high CC, $IE \approx 0$
R2	Bounded	Conservative	Moderate RIR, mid CC, modest IE
R3	Unbounded	Aggressive	Internal undecidability, but strongly reduced visibility at the interface;
R4	Unbounded	Conservative	High RIR, non–trivial IE, sustained macro–structure

Regime R1 serves as a non–proto baseline: collapse overwhelms potential complexity. R2 captures architectures where some richness survives but no systematic attempt is made to surface logical limits. R3 corresponds to systems with powerful internal computation whose limits are mostly hidden by downstream collapse. R4 is the regime of interest for *epistemic* architectures: internal limits become observable through the epistemic head and are mirrored by complex CA behaviour under Φ.

5.3 Reticular–CA correspondence and PCE

We can now formulate the central conjecture that links CRC reticula, ECNN diagnostics and CA dynamics, in the spirit of PCE [Wolfram, 1984, Wolfram, 2002].

Conjecture 5.1 (Reticular–CA correspondence under CRC)**.** *Let R be a compact reticulum with at least one Turing–like level, and let $\Phi(R)$ be a one–dimensional CA encoding constructed as in Section 4.3. Then:*

1. *If R exhibits high reticular irreducibility $RIR \approx 1$ on a representative family of traces and admits at least one link $L_i \to L_j$ with $CC_{i\to j} < 1$ and $IE_{i\to j} > 0$, then the CA $\Phi(R)$ displays complex (class IV/V–like) dynamics in the sense of Wolfram's empirical classification [Wolfram, 1984, Li & Packard, 1990].*

2. *Conversely, if a CA rule f yields sustained complex dynamics (class IV/V) on a non–negligible set of initial conditions, then the associated reticulum $\Psi(CA)$ contains a Turing–like level and exhibits high RIR and non–trivial IE for at least one coarse–graining [Cook, 2004, Shalizi & Crutchfield, 2001].*

This is a deliberately *epistemic* version of PCE: it does not state that "most systems in nature are computation–universal", but that, conditional on CRC and the diagnostics (CC, IE, RIR), reticula and CAs occupy correspondingly complex regimes. Any counterexample—for instance, a reticulum with high RIR and non–trivial IE whose $\Phi(R)$ is provably class I—would refute the conjecture.

5.4 Illustrative scenario (conceptual)

Consider an ECNN whose internal architecture includes:

- a differentiable stack or tape that can emulate a universal Turing machine (activated memory);

- a sequence of convolutional and pooling layers with residual paths that preserve quasi–injective channels from the Turing–like core to the epistemic head;
- an epistemic head trained on Popper–χ suites to minimise CR and maximise IR.

If collapse is conservative along at least one path ($CC_{i\to j}^{(\mathrm{ECNN})}$ moderate, not maximal) and $RIR^{(\mathrm{ECNN})}$ is empirically high on relevant tasks, we expect this architecture to fall in regime R4. Under a faithful encoding Φ, its CA image should exhibit complex spatio–temporal structures (gliders, long–lived domains) rather than quick convergence to fixed or purely chaotic behaviour [Wolfram, 1984, Wolfram, 2002].

If the same Turing–like core is followed by aggressive pooling and narrow bottlenecks with no residual bypass, collapse becomes strong ($CC^{(\mathrm{ECNN})} \to 1$ on critical traces), undecidability is largely blocked, and the architecture migrates toward regime R1 or R2. The corresponding CA may then appear class I/II or III despite the latent universality of the underlying core [Li & Packard, 1990]. This illustrates why, in the RLA/CRC view, topology and transmission types, not sheer parameter count, are the primary levers for proto–epistemic behaviour.

6 Roadmap, Threats to Validity and Ethical Note

6.1 Roadmap for empirical and mathematical work

The conceptual bridge proposed here suggests several concrete steps, in line with existing work on deep learning, CA and computational mechanics [Goodfellow et al., 2016, Wolfram, 2002, Shalizi & Crutchfield, 2001]:

1. **Canonical ECNN benchmarks**: define standard Popper–χ suites (contradictory pairs, undecidability triggers, semantic adversaries) for vision and language tasks, and report $(\mathrm{CR}, \mathrm{IR})$ alongside accuracy and calibration.
2. **Reticular diagnostics**: implement practical estimators for $CC^{(\mathrm{ECNN})}$, $IE^{(\mathrm{ECNN})}$ and $RIR^{(\mathrm{ECNN})}$ on trained networks, and correlate them with architectural choices (depth, pooling strategy, residual topology, memory).
3. **CA embeddings**: construct explicit encodings Φ and Ψ for small reticula and CA rules, and empirically verify or falsify parts of Conjecture 5.1.
4. **Proto–epistemic design patterns**: systematise architectural motifs (e.g. weak collapse on Turing–critical channels, explicit epistemic heads) that consistently yield low CR, high IR and high RIR, and study their CA images.

6.2 Threats to validity

Several limitations and risks should be emphasised [Popper, 1959, Shalizi & Crutchfield, 2001, Wolfram, 2002]:

- **Approximate diagnostics**: in practice, CC, IE and RIR are estimated on finite samples and restricted probe families. A low estimated IE does not prove absence of emergence; a high RIR may depend on the chosen horizon T.

- **Encoding dependence**: the behaviour of $\Phi(R)$ and $\Psi(\mathrm{CA})$ depends on encoding choices. Poorly chosen encodings can artificially inflate or suppress apparent complexity.

- **Heuristic CA classes**: Wolfram's four classes are empirically useful but coarse. More refined tools (entropy rates, causal states, computational mechanics) should be used in parallel [Shalizi & Crutchfield, 2001].

- **Epistemic horizon choice**: compactness, CRC and the meaning of "emergent" are all relative to the chosen horizon $\mathcal{H}$. Different communities may adopt different horizons for the same underlying system [Kuhn, 1970].

6.3 Ethical note

Epistemic heads and challenge suites are attractive for high–stakes applications (law, medicine, finance) because they can, in principle, make ignorance and contradiction explicit. This also creates new failure modes: over–trust in the epistemic channel, adversarial manipulation of challenge sets, and opaque epistemic matrices. Any practical deployment of ECNNs as decision aids should:

- adopt conservative thresholds for accepting predictions, favouring abstention ($\perp$) when in doubt;

- subject epistemic matrices, reticular designs and Popper–χ suites to independent audit [Popper, 1959];

- clearly separate epistemic outputs (what the model can justifiably claim) from ontological statements about the world.

7 Conclusion

This annex has tightened the structural and conceptual links between:

- the RLA/CRC framework of Annex A [Church, 1936, Turing, 1936/1937, Rice, 1953, Anderson, 1972, Noble, 2012];

- the ECNN architectures and epistemic heads of Annex C [Goodfellow et al., 2016];

- and the CA–based explorations of complexity and universality associated with PCE [Wolfram, 1984, Wolfram, 2002, Cook, 2004].

By resolving the main structural gaps—terminological alignment, explicit reticular diagnostics, and a constructive (if high–level) mapping to CA—the annex turns what could have remained a loose analogy into a falsifiable research programme. The programme is modest in its metaphysical commitments: it does not assert that "everything computes the same way", but it does claim that, under CRC and measurable diagnostics, there is a robust correspondence between: (i) how multi–level models distribute injective and non–injective transmissions, and (ii) the kind of dynamical regimes they inhabit when represented as canonical discrete systems.

Whether Conjecture 5.1 ultimately stands or falls is a matter for mathematical proof and empirical exploration. Either outcome would be informative: a confirmation would sharpen the relevance of PCE for engineered epistemic architectures; a refutation would reveal new ways in which layered computation can escape CA–style universality. In both cases, the reticular perspective provides a disciplined lens for thinking about computation, emergence and the limits of knowledge in complex systems.

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